🌀 3D Coordinates Extension — The Loop Pattern (Beginner)

Topic: One loop, three counters (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Beginner, after Lesson 6) · **Time:** about 30 minutes

Until now you placed blocks by choosing **every** coordinate **by hand**. This time **one loop** builds all **7** obsidian walls for you. The loop keeps three counters — **x**, **y**, **z** — and changes each one a little every pass. That is an **algorithm**: a short list of steps a computer repeats. Load the world, type `SPIRAL` in the chat, and watch 7 walls grow into a pattern.

```
x = 14
y = 0
z = 2
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
    z = z + 2
```

> 🔁 **Big idea:** the loop does the same three changes every pass — x goes **down 2**, y goes **up 1**, z goes **up 2**. So each new wall is a little **wider**, **taller**, and **deeper** than the last.
>
> 🔴 Stand on your **home spot** (red block) every run.
>
> 🎨 Every wall is **obsidian**.

1 · Predict 🔮

Read the code. Circle one.

`for step in range(7)` **— how many walls does the loop build? Circle one:** 3 · 7 · 10

Each pass runs `y = y + 1`**. Does that make the next wall…? Circle one:** taller · wider · deeper

Each pass runs `z = z + 2`**. Does that make the next wall…? Circle one:** taller · wider · deeper

The loop starts with z = 2. After one pass, `z = z + 2` **runs. What is z now? Circle one:** 3 · 4 · 6

2 · Spot the Bug 🐛

The `z = z + 2` line was deleted. Now z stays **2** every pass, so all 7 walls land in the **same spot**, on top of each other — you see one wall, not a pattern.

```
for step in range(7):
    fill OBSIDIAN from (14, 0, z) to (x, y, z)
    x = x - 2
    y = y + 1
```

What is missing? Circle one: the z step · the loop

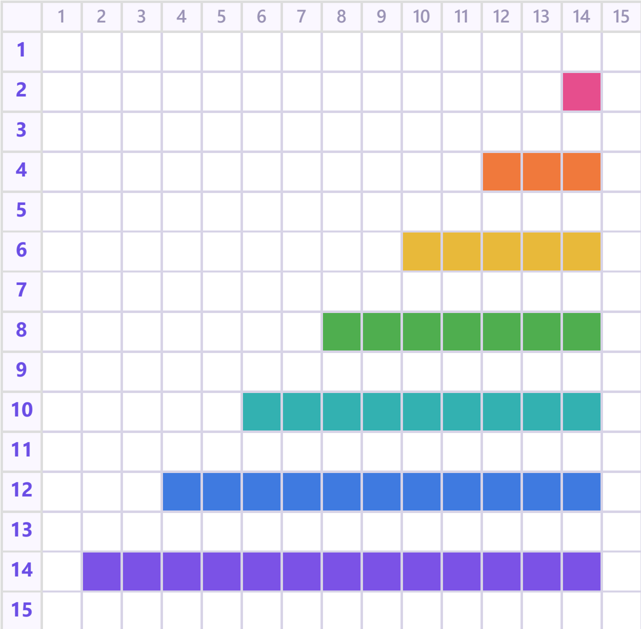
Should each wall sit at a new z or the same z? Circle one: a new z · the same z

3 · Show Your Work 📸 🎥

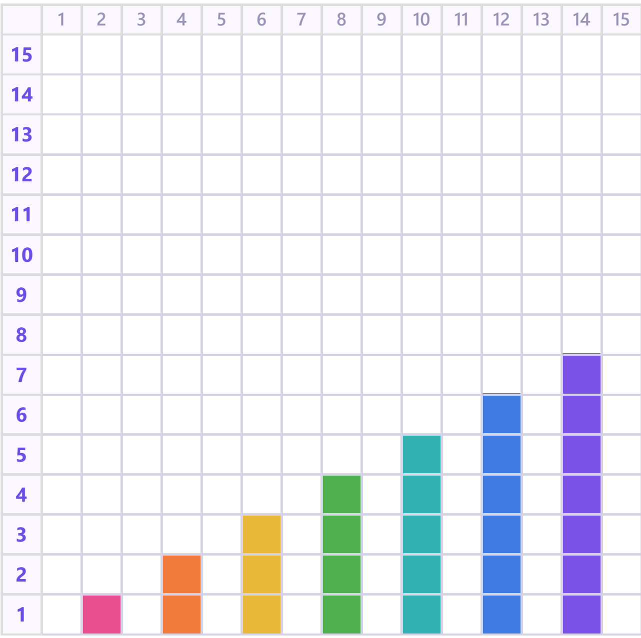
Load the world. Stand on your home spot. Type `SPIRAL` in the chat and watch the loop build all 7 walls. Walk around the pattern.

🧱 **Same build, two views. Colour = the z value.**

👁 **Top view — looking DOWN (x → across, z ↓ deeper)**



↔️ **Side view (z → deeper, y ↑ up)**



colour = z value: 🟥 2 🟧 4 🟨 6 🟩 8 🟦 10 🟦 12 🟪 14

Reading left to right, the steps get taller: 1, 2, 3, 4, 5, 6, 7.

How many steps are there? Circle one: 5 · 7 · 9

Which step is the tallest? Circle one: the first (front) · the last (back)

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera at the pattern. Name the smallest and biggest step.

Say these out loud, filling in the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.